

Conducting Forensic Investigations on Network Infrastructure (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 09

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Time on Task:

2 hours, 30 minutes

Progress:

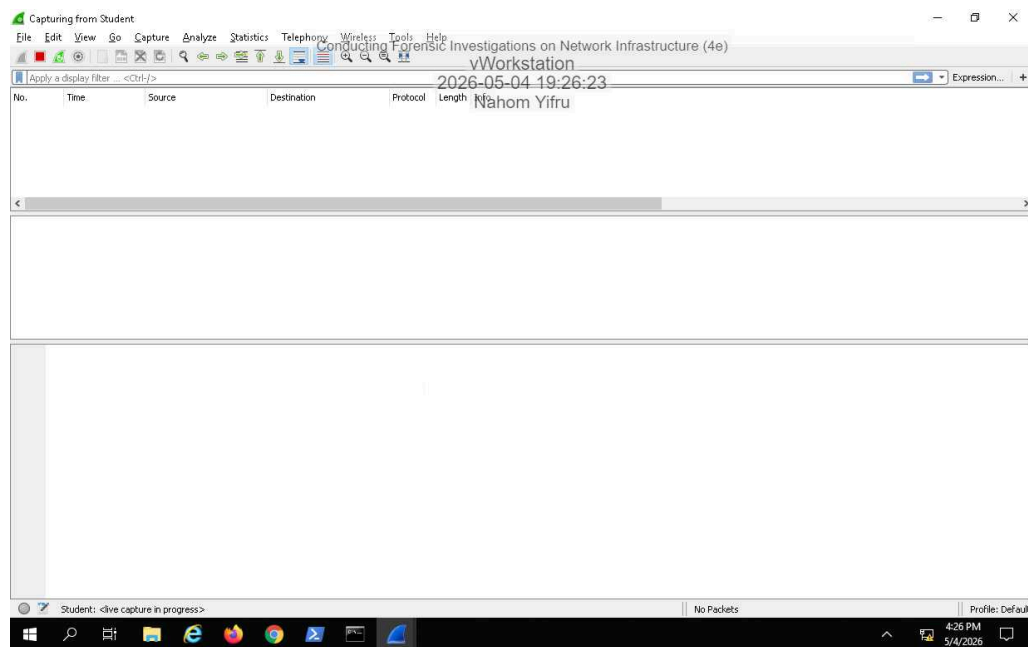
84%

Report Generated: Monday, May 4, 2026 at 7:59 PM

Section 1: Hands-On Demonstration

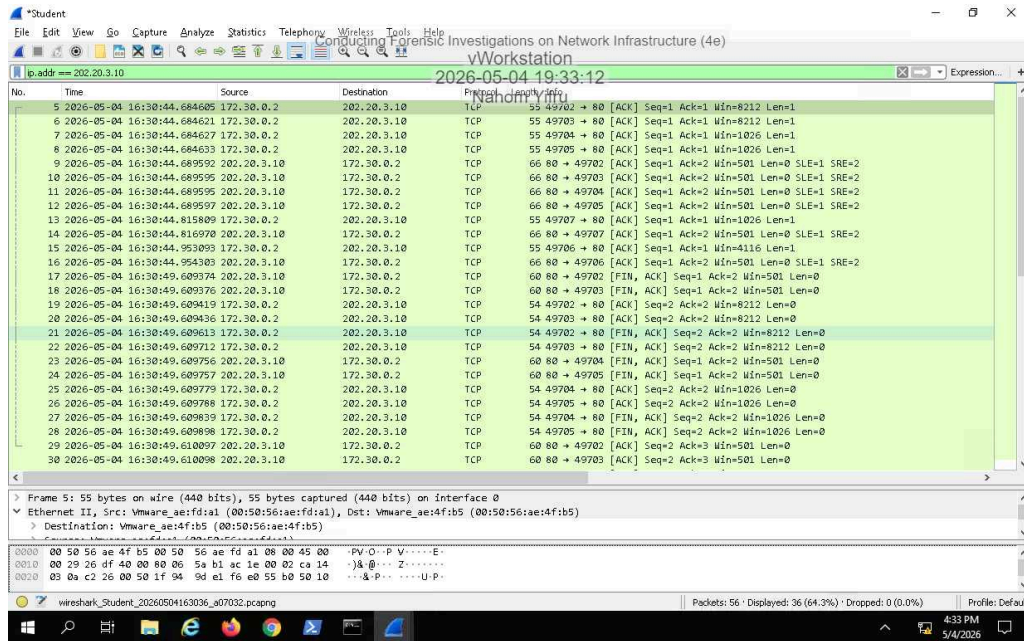
Part 1: Perform Packet Capture and Analysis

11. Make a screen capture showing the **timestamp-sorted traffic**.

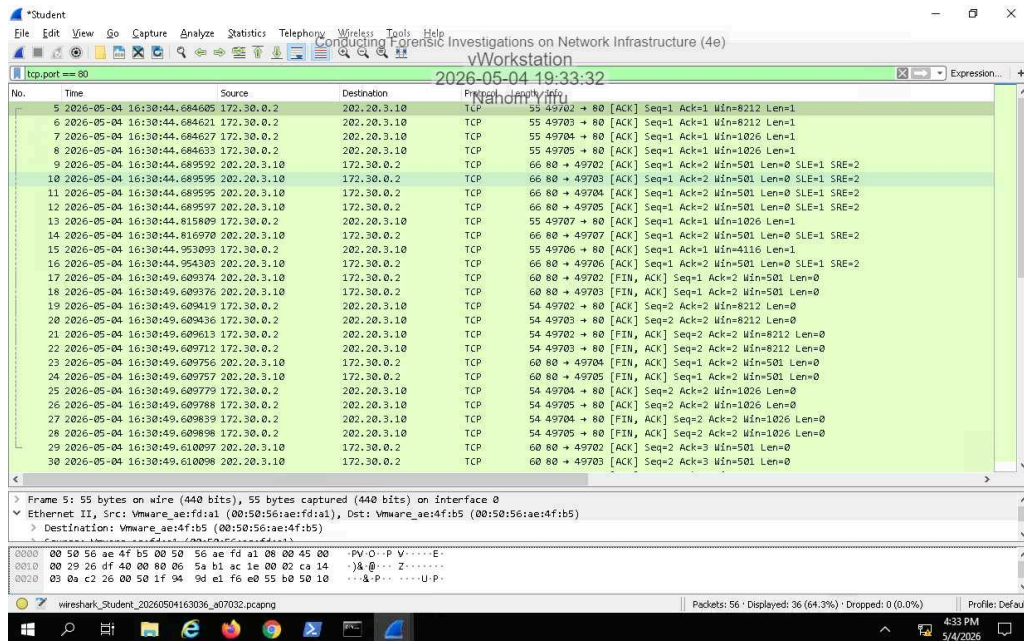


there is no internet to the account idk if I'm supposed to fix that first... (bottom corner)and there is no network and i cant comeplet this assignment.

13. Make a screen capture showing the IP-filtered traffic.



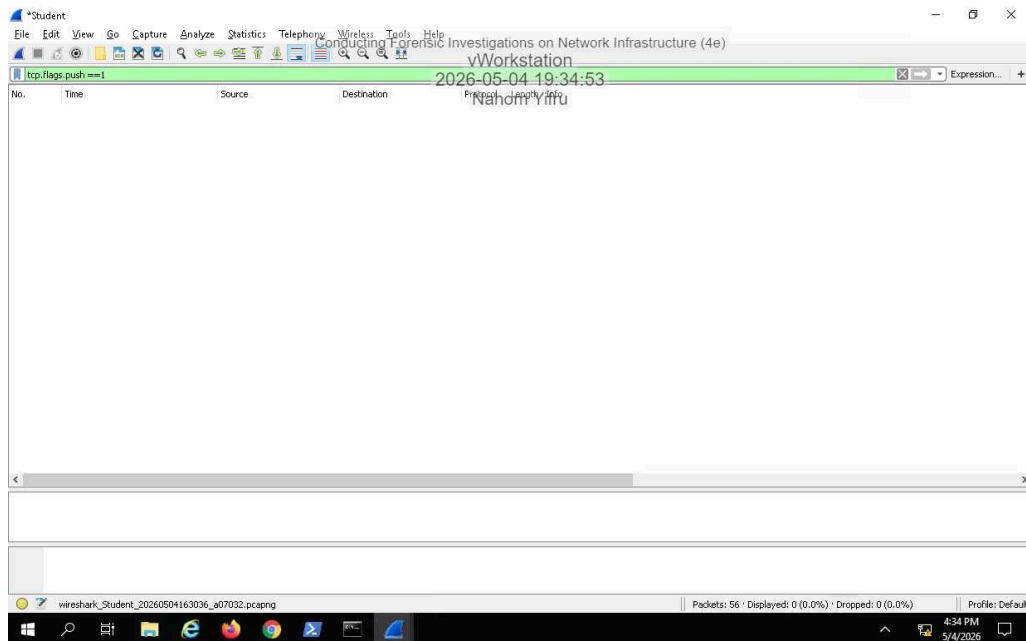
15. Make a screen capture showing the port-filtered traffic.



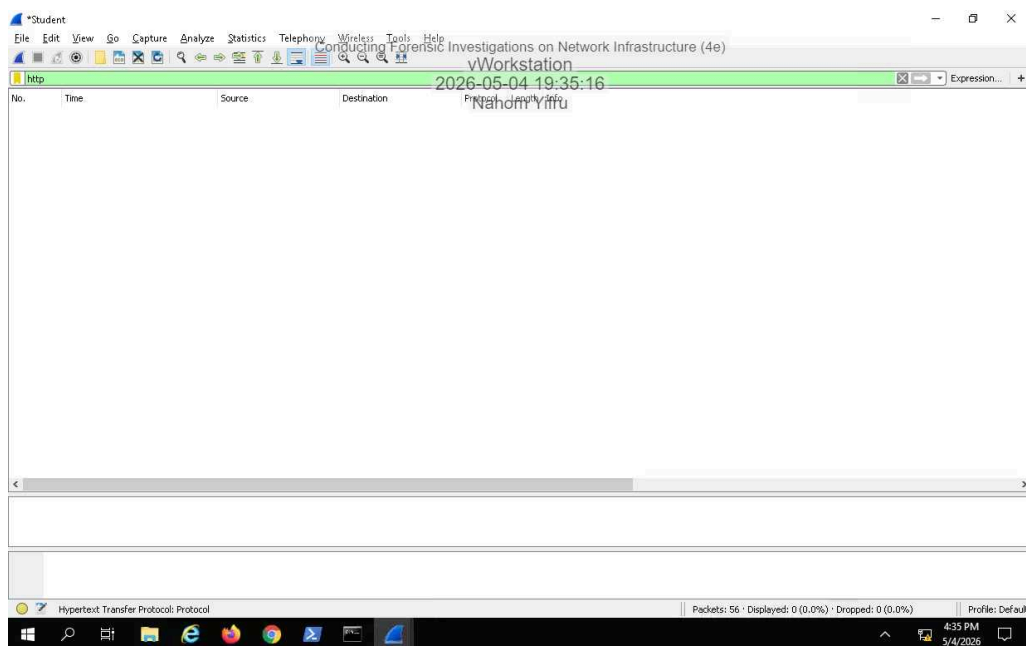
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17. Make a screen capture showing the **TCP push flag-filtered** traffic.



19. Make a screen capture showing the **http-filtered** traffic.



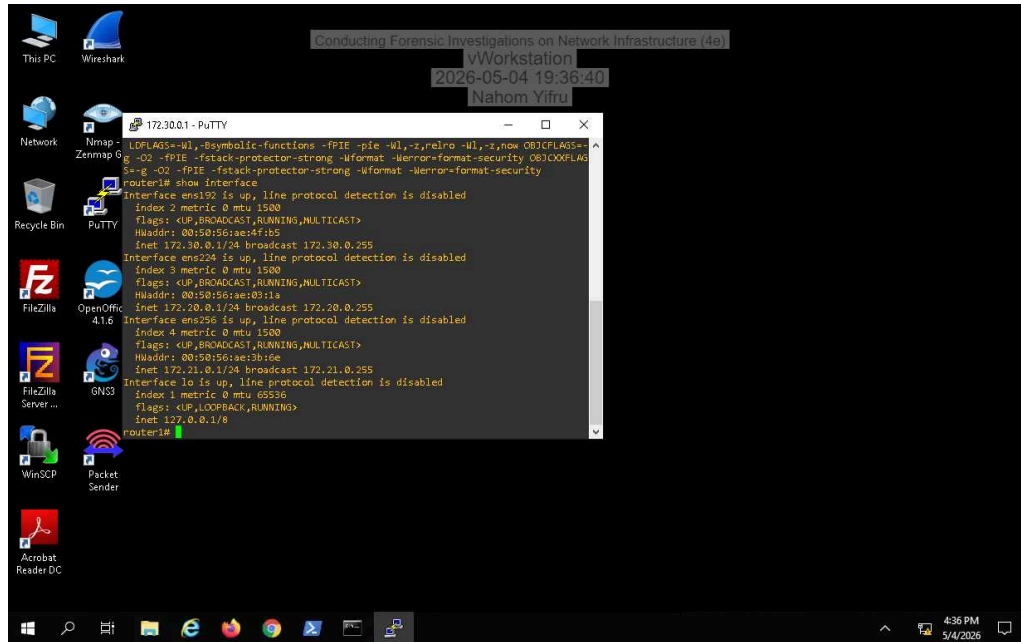
i may have not waited long enough

Part 2: Analyze a Router for Forensic Evidence

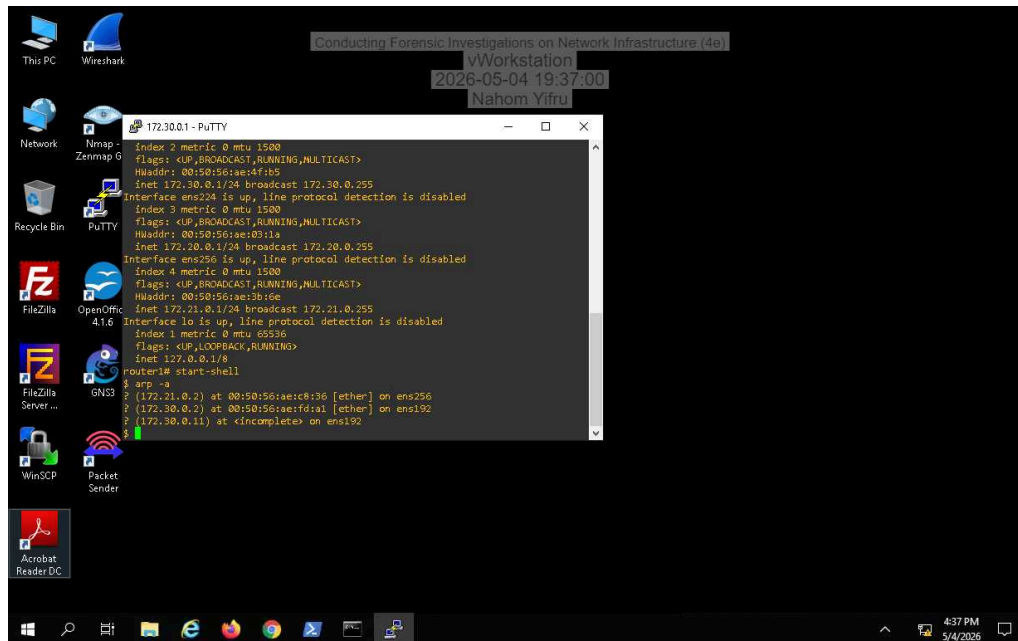
5. Make a screen capture showing the router's version output.

Incomplete

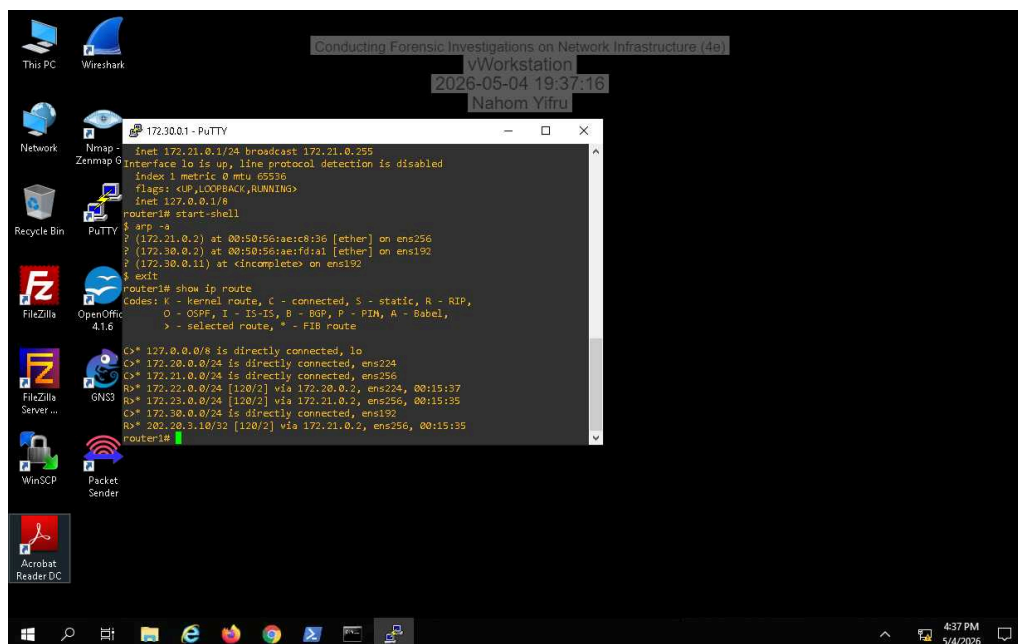
7. Make a screen capture showing the router's interface details.



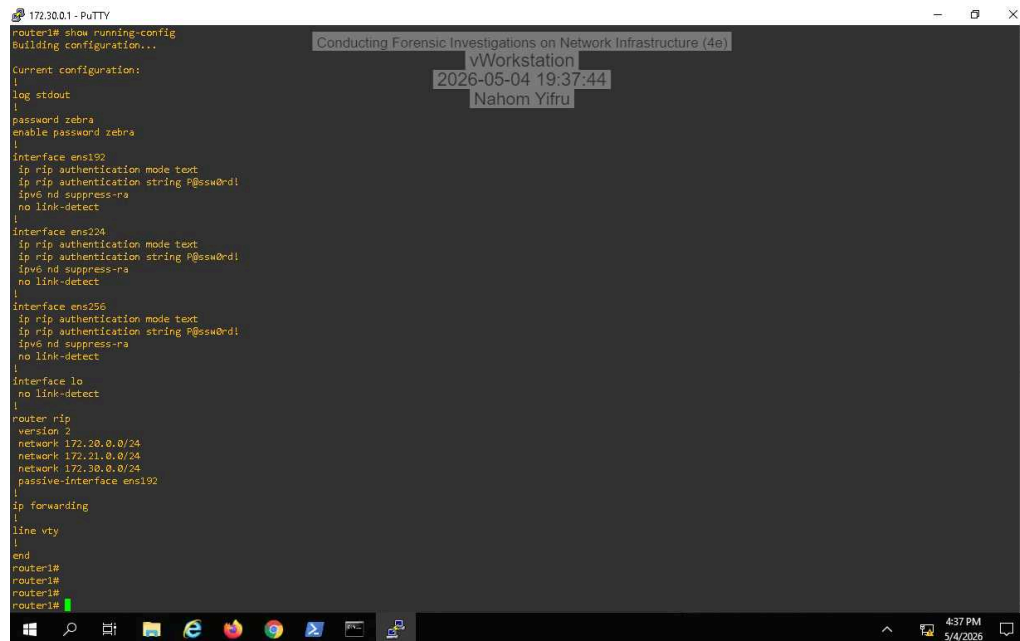
10. Make a screen capture showing the router1 ARP table.



13. Make a screen capture showing the IP routing table.



15. Make a screen capture showing the **currently running configuration**.



The screenshot shows a PuTTY terminal window titled "172.30.0.1 - PuTTY". The terminal displays the output of the command "router1# show running-config". The configuration includes settings for interfaces ens192, ens224, and ens256, all configured with RIP authentication and link detection. It also shows the router configuration for RIP version 2, including network statements and a passive-interface. The terminal output is as follows:

```
router1# show running-config
Building configuration...

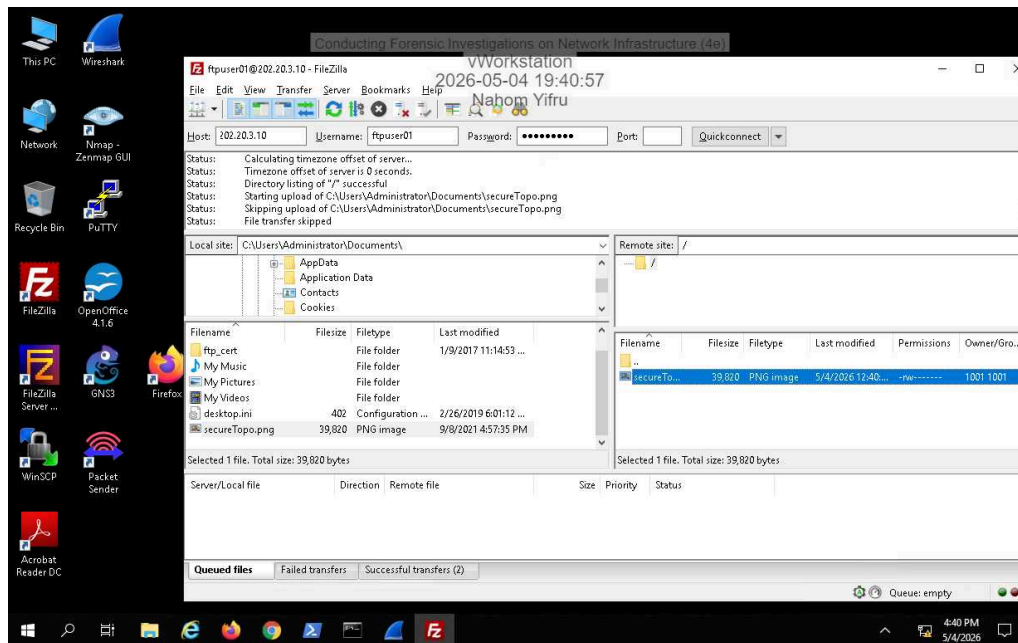
Current configuration:
!
log stdout
!
password zebra
enable password zebra
!
interface ens192
 ip rip authentication mode text
 ip rip authentication string P@ssw0rd!
 ipv6 nd suppress-ra
 no link-detect
!
interface ens224
 ip rip authentication mode text
 ip rip authentication string P@ssw0rd!
 ipv6 nd suppress-ra
 no link-detect
!
interface ens256
 ip rip authentication mode text
 ip rip authentication string P@ssw0rd!
 ipv6 nd suppress-ra
 no link-detect
!
interface lo
 no link-detect
!
router rip
 version 2
 network 172.30.0.0/24
 network 172.21.0.0/24
 network 172.30.0.0/24
 passive-interface ens192
!
 ip forwarding
!
line vty
!
end
router1#
router1#
router1#
```

The terminal window also has a watermark overlay that reads "Conducting Forensic Investigations on Network Infrastructure (4e)", "vWorkstation", "2026-05-04 19:37:44", and "Nahom Yifru". The Windows taskbar at the bottom shows the time as 4:37 PM on 5/4/2026.

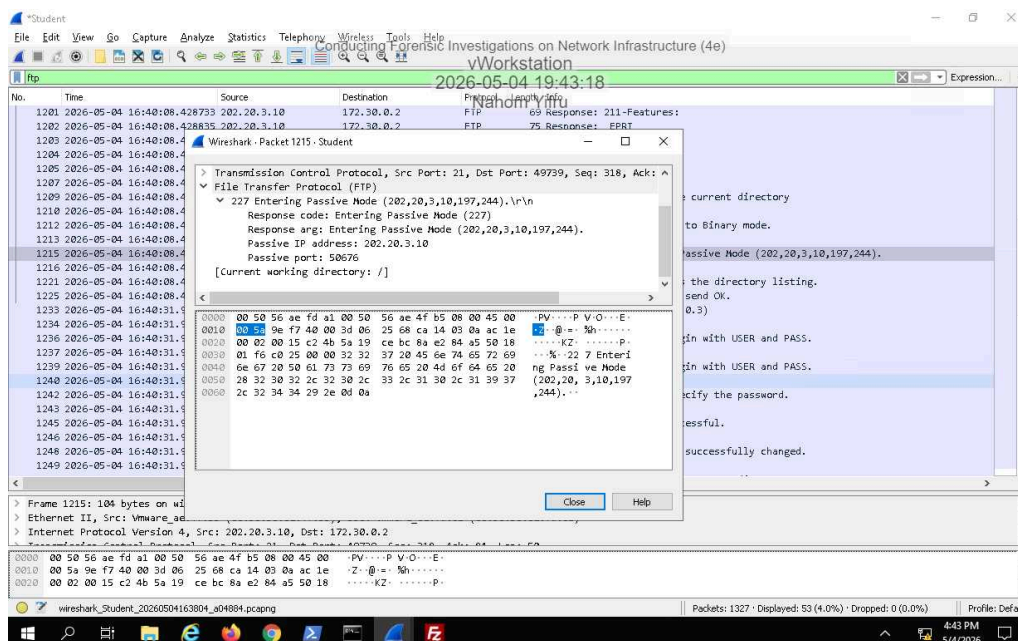
Section 2: Applied Learning

Part 1: Perform Advanced Packet Capture and Analysis

7. Make a screen capture showing the successful transfer of the secureTopo.png file.



15. Make a screen capture showing the passive port specified by the FTP server in the Packet Details pane.



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The screenshot displays the Wireshark 2.6.2 interface with a packet capture of an IP address 202.20.3.10. The packet list shows a SYN packet (Seq=8761) and an ACK packet (Seq=10221). The packet details pane shows the structure of the Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol layers. The packet bytes pane shows the raw data in hexadecimal and ASCII.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
49	2026-05-04 16:38:20.8	16:38:20.869887	172.30.0.2	TCP	66	49719 → 80 [SYN, ECN, CWR] Seq=0
50	2026-05-04 16:38:20.8	16:38:20.870289	202.20.3.10	TCP	1514	80 → 49718 [ACK] Seq=8761 Ack=349
51	2026-05-04 16:38:20.8	16:38:20.870290	202.20.3.10	TCP	1514	80 → 49718 [ACK] Seq=8761 Ack=349
52	2026-05-04 16:38:20.8	16:38:20.870291	202.20.3.10	TCP	1514	80 → 49718 [ACK] Seq=8761 Ack=349

Packet Details:

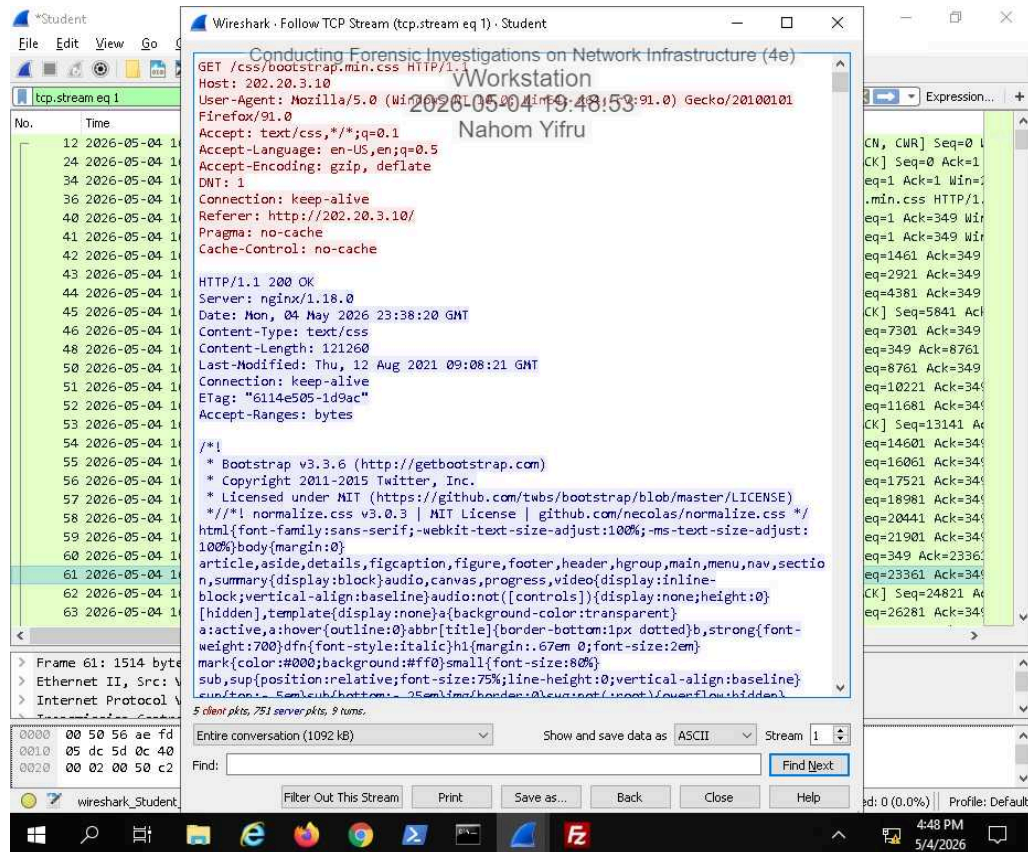
- Frame 61: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on
- Ethernet II, Src: Vmware_00:50:56:ae:4f:b5, Dst: Vmware_00:50:56:ae:4f:b5
- Internet Protocol Version 4, Src: 202.20.3.10, Dst: 172.30.0.2
- Transmission Control Protocol, Src Port: 80, Dst Port: 49718, Seq: 23361, Ack:

Packet Bytes:

```

0000  00 50 56 ae fd a1 00 50 56 ae 4f b5 08 00 45 00  .PV....P V.O...E-
0010  05 dc 5d 0c 40 00 3d 06 61 d1 ca 14 03 0a ac 1e  .j|@:~ a.....
0020  00 02 00 50 c2 36 c6 41 5d 94 da b3 b2 f0 50 10  .P-6 A |.....P-
  
```


20. Make a screen capture showing the Follow TCP stream window.



32. Make a screen capture showing the reconstituted PNG file.

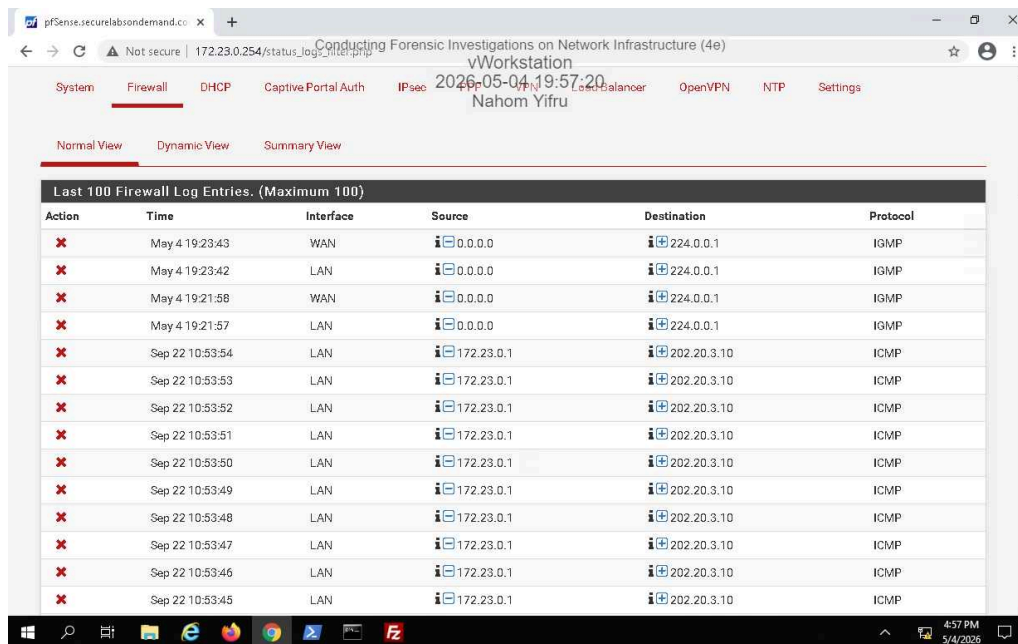
```
File Edit Format View Help
GET /css/bootstrap.min.css HTTP/1.1
Host: 202.20.3.10
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win32; x64; rv:91.0) Firefox/91.0
Accept: text/css,*/*;q=0.1
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
DNT: 1
Connection: keep-alive
Referer: http://202.20.3.10/
Pragma: no-cache
Cache-Control: no-cache

HTTP/1.1 200 OK
Server: nginx/1.18.0
Date: Mon, 04 May 2026 23:38:20 GMT
Content-Type: text/css
Content-Length: 121260
Last-Modified: Thu, 12 Aug 2021 09:08:21 GMT
Connection: keep-alive
ETag: "6114e505-1d9ac"
Accept-Ranges: bytes

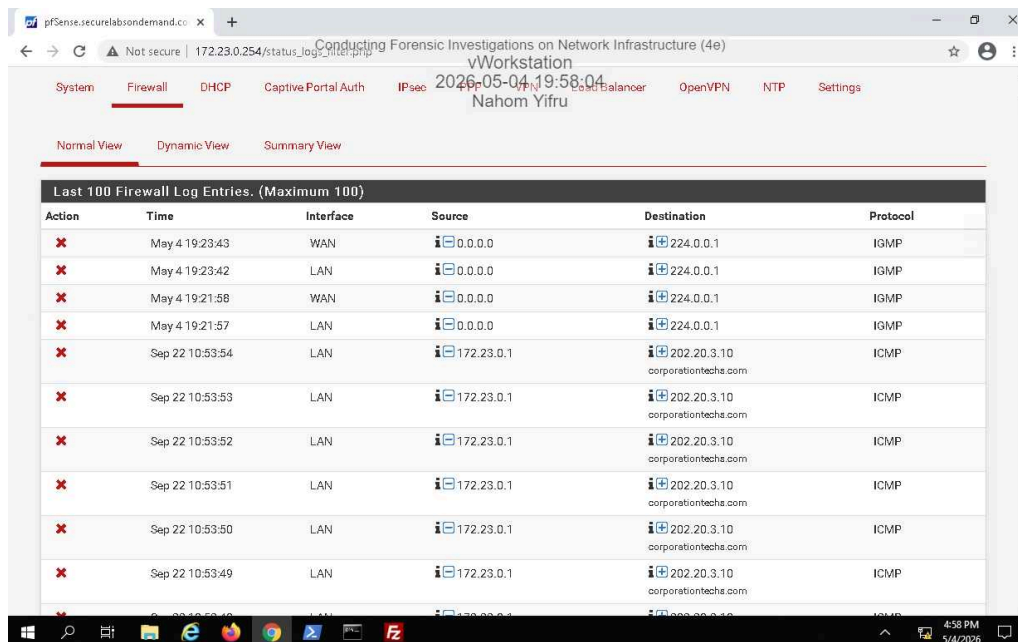
/*!
 * Bootstrap v3.3.6 (http://getbootstrap.com)
 * Copyright 2011-2015 Twitter, Inc.
 * Licensed under MIT (https://github.com/twbs/bootstrap/blob/master/LICENSE)
 */
/*! normalize.css v3.0.3 | MIT License | github.com/necolas/normalize.css */
html{font-family:sans-serif;-webkit-tgroup,select,textarea{margin:0;font:inherit;color:inherit}button{overflow:visible}button,select{text-transform:n
order-collapse:collapse}td,th{padding:0}/*! Source: https://github.com/h5bp/html5-boilerplate/blob/master/src/css
s/glyphicons-halflings-regular.eot);src:url(../fonts/glyphicons-halflings-regular.eot?#iefix) format('embedded-op
re{content:"\e003"}.glyphicon-heart:before{content:"\e005"}.glyphicon-star:before{content:"\e006"}.glyphicon-star
ent:"\e029"}.glyphicon-repeat:before{content:"\e030"}.glyphicon-refresh:before{content:"\e031"}.glyphicon-list-al
yphicon-align-right:before{content:"\e054"}.glyphicon-align-justify:before{content:"\e055"}.glyphicon-list:before
e078"}.glyphicon-chevron-left:before{content:"\e079"}.glyphicon-chevron-right:before{content:"\e080"}.glyphicon-p
ent:"\e104"}.glyphicon-eye-open:before{content:"\e105"}.glyphicon-eye-close:before{content:"\e106"}.glyphicon-war
\e127"}.glyphicon-hand-left:before{content:"\e128"}.glyphicon-hand-up:before{content:"\e129"}.glyphicon-hand-down
>
```

Part 2: Analyze a Firewall for Forensic Evidence

9. Make a screen capture showing the entries in the firewall log.



11. Make a screen capture showing the resolved entries in the firewall log.



Section 3: Challenge and Analysis

Part 1: Identify the Source of a Suspicious Route

Make a screen capture showing the **non-RIP** route that you discovered on the target router.

Incomplete

Part 2: Identify Suspicious Outgoing Connections

Record the destination IP address and Port number of the outgoing connection attempt.

Incomplete